Week 7. By Dery Yufan.

Prob 1. Let D-SAT be the problem that acquire 2 solutions for SAT. Prove D-SAT is NPC.

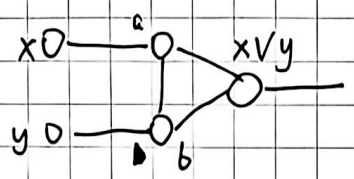
We prove SAT $\le_p$ D-SAT.

For arbitary CNF formula φ, if the virables are $x_1, x_2, \cdots, x_n$. We construct φ' as: $\varphi' = (\varphi) \wedge (x_{n+1} \vee \neg x_{n+1})$. (thus there are $n+1$ virables in φ').

Easy to see $\vec{x}$ is a sol. for φ iff $\vec{x} \cup x_{n+1}$ and $\vec{x} \cup \neg x_{n+1}$ are sol. for φ'.

Thus SAT $\le_p$ D-SAT. Since SAT is NPC. D-SAT is NPC.

Prob 2. a. Consider 3-COLORING problem. and consider following figure. Let colors are $\{T, F, B\}$. Prove the gadget works like OR gate.



gadget.

① If $x = y = F$. then $a \ne F$. $b \ne F$. $a \ne b$. thus $\{a, b\} = \{T, B\}$.

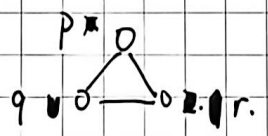
Since $x \vee y \ne a$. $x \vee y \ne b$. so $x \vee y = F$.

② If $x = T$ ($y = T$ is likewise). then $a \ne T$. Whatever y is. we can let $\begin{cases} a = F \\ b = B \\ x \vee y = T \end{cases}$.

and it's a valid solution. ~~and if $x \vee y = F$, then $a = B$, $b = T$~~

b. Prove 3-COLORING is NPC.

Consider a gadget called initial:



Obviously $p \ne q$. $q \ne r$. $r \ne p$.

so we can suppose $p = T$. $q = F$. $r = B$. We prove SAT $\le_p$ 3-COLORING by construct a graph.

For arbitary CNF formula φ in SAT prob., assume virables are $x_1, x_2, \cdots, x_n$.

~~build n vertices~~

The graph contains initial uniquely. then add $2n$ vertice taken for $x_1, \cdots, x_n, \neg x_1, \cdots, \neg x_n$. ~~$x_i$ is connected with r, B, and n vertices taken~~ x_i. $\neg x_i$, $r = B$ connect as 3-vertice ring.

For every clause, we consider the or oper. one by one. For every or oper., we take the last result (if it is the first oper., take the left hand virable), and the right hand virable. (x_i or $\neg x_i$. depend on the clause itself). through the or gate. To make sure the output is T or F, we connect the output vertice with $r = B$.

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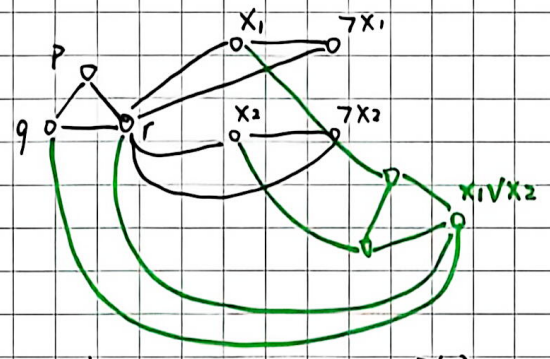
(Ones might concern that TVF may be F. but we forced the whole result to be T. so F is worse than T)

Then for all clause we can calc. the result. For all results of clauses, to make sure the result is T. we connect the result vertice with $q = F$.

Consider the graph now. It has a 3-coloring iff the formula has solution. thus SAT $\leq_p$ 3-COLORING. and 3-COLORING is NPC.



A sample figure for formula $(x_1 \vee x_2)$.

Prob 3. Show $Z(n)$ and $n^{Z(n)}$ are computable in poly of n. $Z(n)$ is the func. in Ladner's thm.

Suppose it take $T(n)$ to calculate $Z(n)$. We have

$$T(n) = T(n-1) + O(|\Sigma|^{\log n} \cdot (\log n)^{Z(n-1)}) + O(n)$$

(Calc $\log n$ and $\lg(\lg n)$.)

(calc $Z(n-1)$). (for all input x. shorter than $\log n$, run $|x|^{Z(n-1)}$ steps on $M_{Z(n-1)}$).

$$= T(n-1) + O(n^{\log|\Sigma|} \cdot (\lg n)^{\log\lg n}) + O(n).$$

Since $(\log n)^{\log\log n} = e^{(\log\log n)^2}$, and $(\lg\lg n)^2 < c\log n \ \forall c$.

$[(\lg\lg n)^2]' = 2 \cdot \frac{1}{n} \cdot \frac{\lg\lg n}{\log n}$. $[c\lg n]' = c \cdot \frac{1}{n}$. and $\frac{\lg\lg n}{\lg n} < \varepsilon$.

$[\lg\lg n]' = \frac{1}{n\lg n}$. $[\varepsilon\lg n]' = \frac{\varepsilon}{n}$. $\frac{1}{\log n} < \varepsilon$.

So $(\log n)^{\lg\lg n} < e^{\varepsilon\lg n} = n^{\varepsilon}$.

$$T(n) = T(n-1) + O(n^{\lg|\Sigma| + \varepsilon + 1}).$$

$$= O(n \cdot n^{\lg|\Sigma| + \varepsilon + 1})$$

$$= O(n^{\lg|\Sigma| + 2 + \varepsilon}) \in P.$$

And $2^{Z(n)} = O(2^{\lg\lg n}) = O(\lg n) \in P$.

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